

Spatial integration of sensory input and motor output in *Pseudomonas aeruginosa* chemotaxis through colocalized distribution

Zhengyu Wu, Maojin Tian, Sanyuan Fu, Min Chen, Rongjing Zhang , Junhua Yuan 

Research Center of Translational Medicine, Central Hospital Affiliated to Shandong First Medical University, Jinan, Shandong 250013, China • Hefei National Research Center for Physical Sciences at the Microscale and Department of Physics, University of Science and Technology of China, Hefei, Anhui 230026, China • Research Center of Translational Medicine, Jinan Central Hospital, Shandong University, Jinan, Shandong 250013, China • Center of Translational Medicine, Zibo Central Hospital Affiliated to Binzhou Medical University, Zibo, Shandong 255036, China

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Abstract

The opportunistic pathogen *Pseudomonas aeruginosa* serves as a model organism for studying multiple signal transduction pathways. The chemoreceptor cluster, a core component of the chemotaxis pathway, is assembled from hundreds of proteins. The unipolar distribution of receptor clusters has long been recognized, yet the precise mechanism governing their assembly remains elusive. Here, we directly observed the relative positions of the flagellar motor and chemoreceptor cluster using flagellar filament labeling and gene editing techniques. Surprisingly, we found that both are located at the same cell pole, with the distribution pattern controlled by the polar anchor protein FlhF. Additionally, the efficient assembly of the chemoreceptor cluster is partially dependent on the integrity of the motor structure. Furthermore, we discovered that overexpression of the chemotaxis regulatory protein CheY leads to high intracellular levels of the second messenger c-di-GMP, triggering cell aggregation. Therefore, the colocalization of the chemoreceptor cluster and flagellum in *P. aeruginosa* serves to avoid cross-pathway signaling interference, enabling cells to conduct various physiological activities in an orderly manner.

eLife assessment

This **valuable** study by Wu et al presents data on bacterial cell organization, with the goal of demonstrating that the two structures that account for bacterial motility - the chemotaxis complex and the flagella - colocalize to the same pole in *Pseudomonas aeruginosa* cells and expose the regulation underlying their spatial organization and functioning. The subject is of importance and the manuscript is well written. However, the work is **incomplete** and the conclusions are too strong for the presented data. This manuscript will be of interest to cell biologists, mainly those studying bacteria, but not only, if the evidence is substantiated, the assumptions clarified, and the novelty is made clear during the revision process.

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Introduction

In biological systems, the process by which proteins self-assemble into organized complex structures is ubiquitous^{1,2}. Pole-to-pole protein oscillations in the *Min* system ensure that the FtsZ ring, a crucial component of cell division, is placed precisely in the middle of the cell body^{3,4}. In response to antibiotic treatment and heat stress, some bacteria generate multiple protein foci throughout the cytoplasm^{5,6}. The polar localization of the secretion system, such as the type VI secretion system, is mediated by targeting proteins and potentially facilitates host-pathogen interactions^{7,8}. Thus, spontaneous assembly is accompanied by the formation of spatial patterns that order the intracellular environment.

To grow and colonize within the host, bacteria have evolved a mechanism for migrating towards more favorable environments. The locomotion organ (the flagellar motor) and the chemotaxis pathway play crucial roles in achieving this goal. The former provides physical drive, while the latter offers directional guidance. The initiation of cell reorientation is controlled by motor switching, which is governed by the chemotaxis two-component signaling pathway. This pathway comprises transmembrane chemoreceptors, a histidine kinase CheA, an adaptor protein CheW, a response regulator CheY, and two adaptation proteins, CheB and CheR. Receptors collaborate with CheA and CheW to form chemosensory arrays in a hexagonal pattern^{9,10}. Upon phosphorylation of CheY by the kinase at the array, it freely diffuses to the motor and binds to FliM, a component of the motor switch complex, to affect motor switching^{11–13}. It is worth noting that there exists diversity among different bacterial species in the placement of flagellar motors¹⁴, whereas chemoreceptor clusters are typically found at one or both ends of the cell¹⁵. These two structural units are functionally connected. However, it remains unclear whether there is an interaction in their spatial distribution and what the specific regulatory mechanism might be.

Pseudomonas aeruginosa, a common human opportunistic pathogen, possesses four chemosensory pathways that perform distinct functions and are stimulated by signal binding to 26 chemoreceptors¹⁶. Among them, F6 pathway-related receptors that mediate chemotaxis, as well as the flagellum are localized at the same cell pole^{17,18}. Furthermore, the polar anchor protein FlhF has been reported to guide the flagellum to grow at the cell pole^{19,20}. However, the mechanism by which chemoreceptor array-related proteins aggregate and self-assemble at the cell pole is controversial. The distribution of the chemosensory arrays in the peritrichous bacterium *Escherichia coli* has been extensively studied and attributed to several mechanisms, including stochastic self-assembly^{21,22}, membrane curvature sorting^{23,24}, and inefficient

clustering in the lateral region²⁵. However, these mechanisms cannot explain the unipolar distribution pattern observed in *P. aeruginosa*. Unlike *E. coli*, specific genes responsible for the placement of the chemosensory array have been identified in several bacterial species. In *Caulobacter crescentus*, chemosensory arrays assemble at the new cell pole with the help of TipN and TipF proteins²⁶. A tripartite ParC-ParP-CheA interaction network was reported to promote polar localization of chemosensory array in *Vibrio parahaemolyticus*^{27, 28}. However, as there are no related genes in *P. aeruginosa*, the mechanism of its chemoreceptor array distribution remains a mystery.

Here, we combined gene editing and in vivo fluorescence imaging of flagellar filaments to directly observe the distribution of chemoreceptor arrays and flagellar motors, proposing a cooperative construction model of chemotaxis network and flagella during the entire division cycle of *P. aeruginosa*. The core focus of this study is to clarify the regulatory mechanism of its chemoreceptor array distribution. We found a substantial association between the assembly of the flagellar motor and the chemoreceptor array. The complete assembly of the motor serves as a partial prerequisite for the assembly of the receptor array, and its assembly site is also regulated by the polar anchor protein FlhF. Furthermore, by introducing exogenously expressed CheY protein, we found that this triggers the expression of c-di-GMP at a high level. From this, we infer that the colocalization of the chemoreceptor array and the flagellum in *P. aeruginosa* avoids the cross-pathway interference of signaling molecules, thus providing a guarantee for the coexistence of multiple chemosensory pathways.

Results

Robust generation of daughter cell with both chemotaxis network and flagellar motor

CheY has been shown to colocalize with chemoreceptors²⁹. To visualize the distribution of chemoreceptor arrays in cells, we fused the gene encoding the enhanced yellow fluorescent protein (eYFP) to the *cheY* gene in the *P. aeruginosa* chromosome (**Fig. 1A**). All genetic modifications were carried out at the native position on the chromosome, allowing the expression of chemotaxis proteins in precise stoichiometry from their natural promoters (**Fig. 1B**). The mutant strain with *cheY-eyfp* was able to moderately expand on soft agar plates¹⁷, proving that its chemotaxis behavior was not notably affected, so it was selected for subsequent experiments. As shown in **Fig. 1C**, CheY-eYFP was mainly located at the cell pole, suggesting that CheA, CheW and CheY in *P. aeruginosa* form a signal transduction complex that was primarily distributed at the cell pole, as described previously¹⁷. From our measurements, nearly 90% (332/372) of cells contain obvious receptor clusters, and factors such as fluorescence bleaching may have caused the loss of fluorescent spots in some cells. In addition, we observed the presence of fluorescent spots at both ends of some cells with a large aspect ratio (probably approaching cell division), indicating that the newly generated progeny cells will have a complete chemotaxis network. Next, we sought to observe the distribution of the flagella in cells simultaneously to understand the motility of future progeny.

Our recent study demonstrated the successful labeling of the flagellar filament with thiol-reactive fluorescent dye by introducing cysteine mutations into the flagellin FliC^{30, 31} (**Fig. 1B**). To avoid fluorescence interference, we utilized a dye with an excitation peak near 568 nm, considering that the excitation peak of eYFP is 513 nm, and the emission peak is 527 nm. We employed a xenon lamp as the excitation light source and switched filters to enable simultaneous fluorescence imaging of flagellar filaments and chemosensory arrays within the same cell. Remarkably, our findings revealed a surprising colocalization of chemosensory arrays and flagellar filaments at the same end of the cell body, and this colocalization was consistent, as shown in **Fig. 1D**. Similar to the chemoreceptor arrays, we observed the phenomenon of double

flagella symbiosis in cells with a large aspect ratio, suggesting that *P. aeruginosa* has evolved a cell division mechanism with precise timing regulation. Before cell division, both poles of the mother cell assembled chemoreceptor arrays and flagellar motors. This ensures that daughter cells possess complete motility and chemotaxis, thereby greatly enhancing the environmental adaptability of the population.

FlhF mediates proper unipolar localization of the chemoreceptor array

Given that the physiological and physical environments of both cell poles are nearly identical, it is unlikely that the unipolar distribution of the chemoreceptor array can be attributed to passive regulatory factors. The number and distribution of flagellar motors and chemoreceptor arrays vary among different bacterial species^{14, 15}, and the localization system for flagella has been well studied. Specifically, the FlhF-FlhG system, discovered in several monotrichous bacterial species, has been shown to control the location and number of flagella^{32, 33}. In *P. aeruginosa*, a knockout of *flhF* leads to mis-localized flagellar assembly^{19, 20}. Considering the consistent colocalization pattern between chemosensory arrays and flagellar motors in *P. aeruginosa*, we speculate that the distribution of receptor clusters is also regulated by similar molecular mechanisms.

To investigate the role of FlhF in the localization of receptor arrays, we constructed a $\Delta flhF$ strain for fluorescence observation. The results revealed that the chemoreceptor arrays no longer grow robustly at the cell pole (**Fig. 2A**), and the assembly positions of the flagellar motor change accordingly, colocalizing with the receptor arrays (**Fig. 2B**). We also quantified the proportion of individuals with obvious receptor clusters, which was more than 80% (182/221), similar to the wild-type strain. These experimental results suggest that the polar anchoring protein FlhF has a minimal impact on the assembly efficiency of *P. aeruginosa*'s chemosensory array, but it strictly controls its unipolar distribution. Furthermore, a consistent colocalization between the flagellar motor and chemosensory array was observed, independent of FlhF. Thus, it is crucial to determine whether there is a causal relationship in the assembly order of these two structural units.

Motor structural integrity is a prerequisite for chemoreceptor self-assembly

Cryo-electron microscopy has successfully revealed the complete flagella structure in various bacterial species. This structure includes multiple components such as the inner-membrane MS ring, the cytoplasmic C ring, and the internal protein secretion system, all of which exhibit strong similarities³⁴. This suggests that the core structure of the flagellar motor is highly conserved. The MS ring, composed of 26 FliF proteins, forms the foundation of the flagellar structure³⁵. In addition to serving as a mounting platform for the C ring, the MS ring also acts as a protective shell for the internal protein secretion machinery³⁶. FliG, the main component protein of the C ring, binds directly to the cytoplasmic surface of FliF in a 1:1 ratio³⁷. Thus, the efficient assembly of the C ring requires preassembly of the MS ring. To observe the distribution of the chemosensory array following the disruption of the C ring and MS ring, we constructed $\Delta fliG$ and $\Delta fliF$ mutants.

We performed fluorescence observation on the $\Delta fliG$ mutant, which has a disrupted flagellar motor C ring. The results showed that the proportion of $\Delta fliG$ cells with obvious chemoreceptor clusters decreased significantly compared to the wild-type cells (59.8%, 140/234), although the chemosensory arrays remained at a single cell pole. We also examined the $\Delta fliF$ mutant, which has a disrupted flagellar motor MS ring. Similar to the $\Delta fliG$ mutant, the proportion of individuals with obvious chemoreceptor clusters decreased significantly (62.5%, 202/323). Subsequently, we constructed a $\Delta flhF\Delta fliF$ mutant and observed that the proportion of individuals with obvious chemoreceptor clusters further decreased (50.7%, 341/672). To ascertain whether it is motor

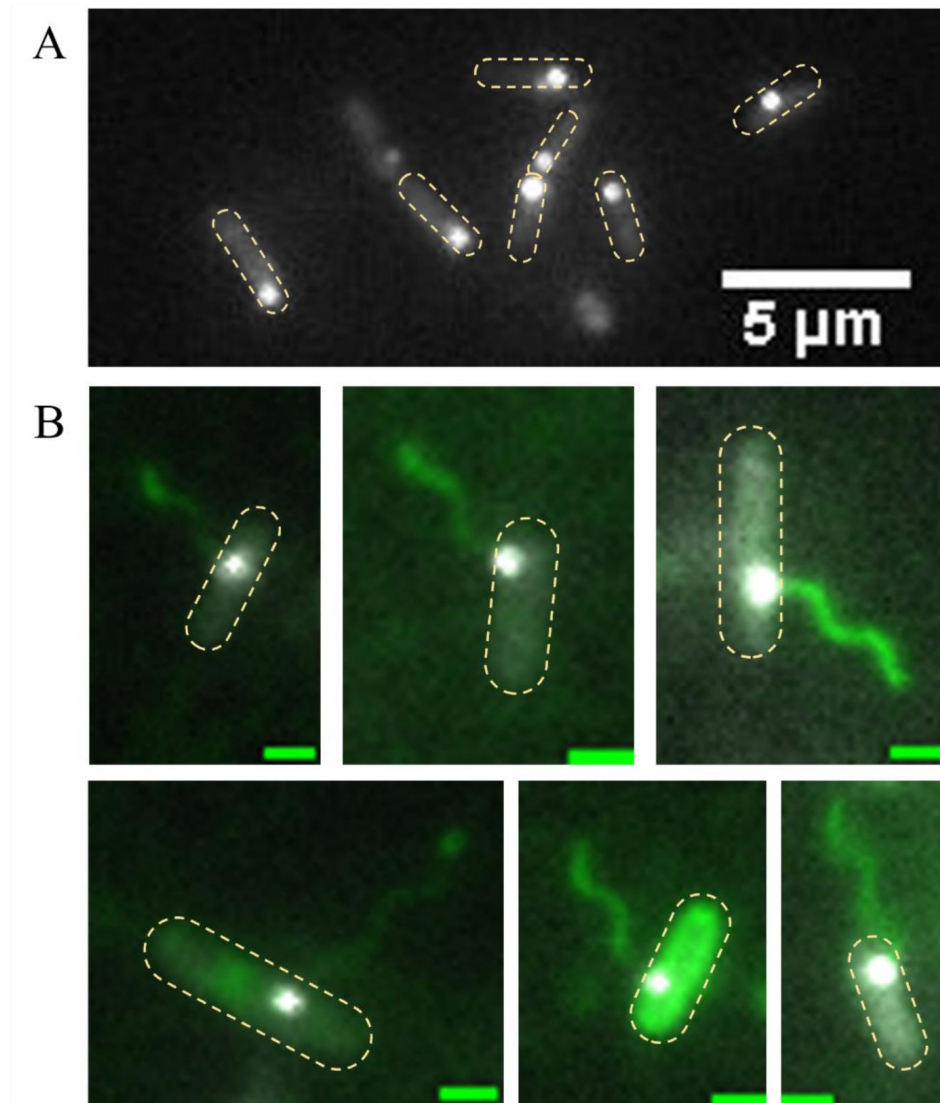


Fig. 2.

A. Subcellular localization of CheY-EYFP in the $\Delta flhF$ strain of *P. aeruginosa*. CheY-EYFP is no longer robustly distributed at the single cell pole. The yellow dashed box marks the cell outline. **B.** The merged imaging of flagellar filaments and CheY-EYFP in the $\Delta flhF$ strain of *P. aeruginosa*, flagellar motor and chemosensory array still colocalize in cells. The yellow dashed box marks the cell outline. The scale bar is 1 μm .

integrity rather than functionality that influences the efficiency of chemosensory array assembly, we constructed a stator mutant ($\Delta\text{motA}\Delta\text{motCD}$). In this mutant, the motor is completely stalled while the structure remains intact. We found that the mutant performed similarly to the wild-type strain in terms of chemosensory array assembly (84.3% of individuals with obvious clusters, 204/242). The fluorescence imaging of receptor clusters for multiple mutants in this section is shown in **Figure 3A**. We utilized Western blotting to measure the expression levels of CheY, which were found to be similar across the different strains (**Fig. 3B**). This further substantiated that the observed phenomenon is based on the structural integrity of the motor rather than the protein expression level. Additionally, we quantified the proportion of cells with receptor clusters and the absolute fluorescence intensity of individual clusters (**Fig. 3C-D**). Overall, our findings suggest that the polar anchor protein FlhF and the structural integrity of the motor are crucial for the formation of chemosensory arrays. The former guides them to the appropriate site, while the latter influences the assembly efficiency of the chemoreceptor complex.

Colocalization of chemosensory array and flagellar motor avoids cross-pathway interference in *P. aeruginosa* signal transduction

The distribution patterns of receptors associated with various signal transduction pathways in *P. aeruginosa* vary considerably. For instance, the biofilm formation-related receptor WspA, unlike the chemotaxis receptors discussed here, is distributed throughout the cell. This distribution is thought to enhance its sensitivity to mechanical perturbations of the cell membrane³⁸. Previous studies speculated that the highly consistent position of chemoreceptor arrays and flagellar motors enhances bacterial chemotaxis performance²⁸. However, a spatial separation between the chemoreceptors and flagellar motors should not significantly impact the temporal comparison in bacterial chemotaxis. The diffusion time of the phosphorylated chemotaxis regulatory protein CheY-P across the longest distance in a bacterial cell body (along the cell's long axis) is approximately 100 ms^{39, 40}, while the timescale for the chemotaxis temporal comparison is on the order of seconds⁴¹. Consequently, the potential significance of the colocalization of chemoreceptors and flagellar motors remains unclear.

The signal transduction pathways in *E. coli* are relatively simple, and the chemotaxis response regulator CheY-P affects only the regulation of motor switching¹¹. In contrast, the intracellular signal transduction pathways in *P. aeruginosa* are more complex. We thus hypothesized that the co-localization of chemoreceptors and flagellar motors in *P. aeruginosa* ensures locally distributed CheY-P molecules, thereby eliminating the need for a high level of intracellular CheY-P and avoiding potential side effects on other signaling pathways.

To test the potential effect of an increased intracellular CheY-P level, we constructed the CheY expression plasmid *cheY*-pJN105, transformed it into the wild-type *P. aeruginosa* strain, and induced it with varying concentrations of arabinose. We observed that the higher the inducer concentration, the more pronounced the cell aggregation became in the field of view (**Fig. 4A**). The transition from planktonic individuals to aggregated communities is similar to biofilm formation, which is known to be accompanied by a significant increase in intracellular c-di-GMP levels.

To investigate whether a higher level of intracellular CheY-P increased the intracellular c-di-GMP level, we sought to detect the c-di-GMP level. We introduced the plasmid pCdrA-gfp into *P. aeruginosa*, using it as a c-di-GMP biosensor with the fluorescence intensity proportional to the c-di-GMP level⁴². To ensure the coexistence of the plasmids, the intracellular CheY concentration was controlled by the plasmid *cheY*-pME6032. We found that the average single-cell fluorescence intensity of the CheY overexpression strain was 71.8% higher than that of the wild-type strain, while the c-di-GMP level of the *cheY* deletion strain was 58.6% lower than that of the wild-type (**Fig. 4B**).

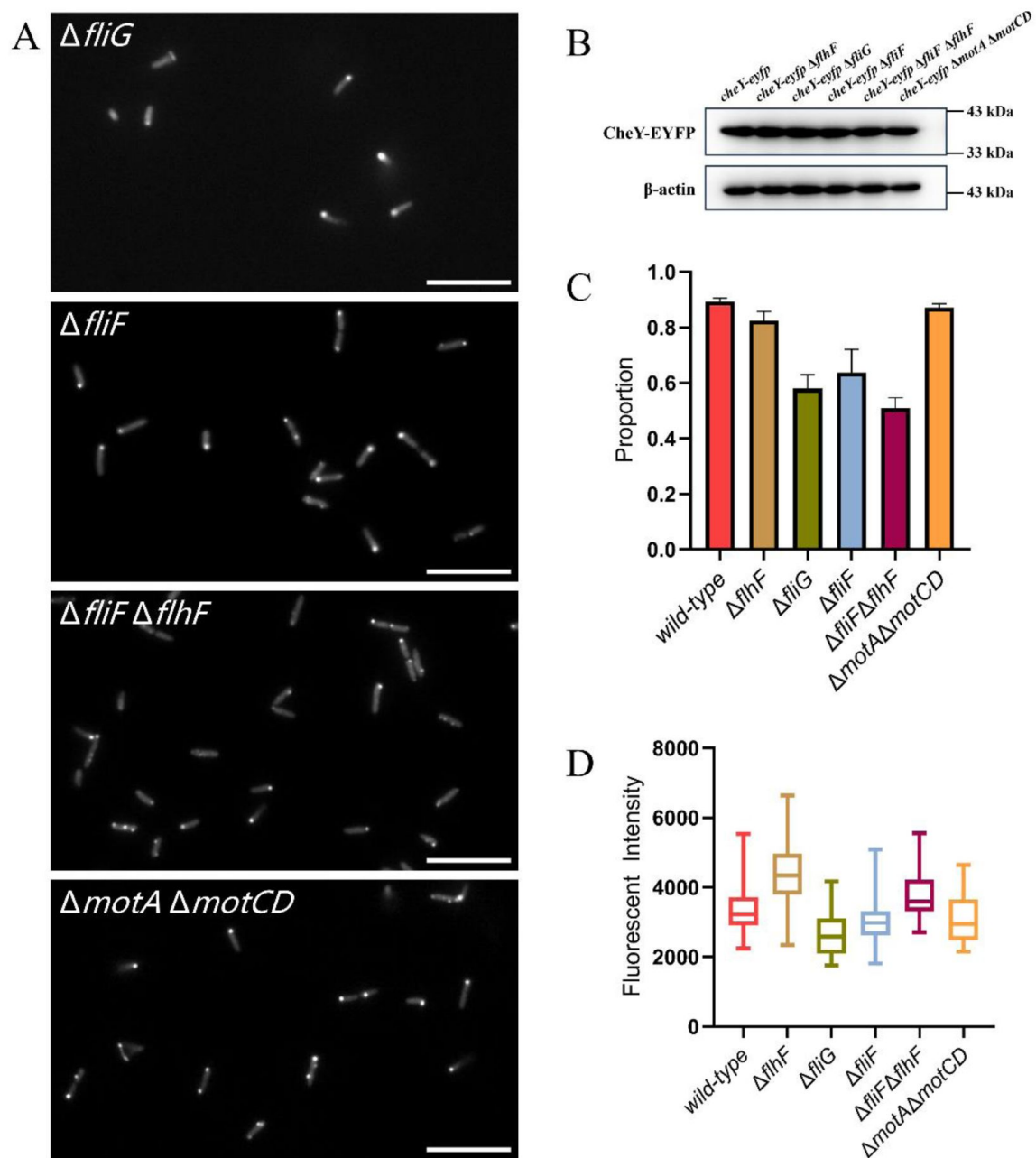


Fig. 3.

A. Subcellular localization of CheY-EYFP in various *P. aeruginosa* strains. The scale bar is 10 μ m. **B.** Western blot analysis was performed to detect CheY expression in various *P. aeruginosa* strains. β -actin was used as the reference protein. **C.** Occurrence probability of obvious chemosensory arrays in *P. aeruginosa* wild-type and several mutant strains. The proportion of individuals with obvious chemosensory arrays decreased significantly in the motor-incomplete strains ($\Delta fliF$ and $\Delta fliG$), and this value was further reduced after *flhF* knockout ($\Delta fliF \Delta flhF$). The $\Delta fliF$ and $\Delta motA \Delta motCD$ strains have a similar chemosensory array occurrence probability as the wild-type strain. **D.** Absolute fluorescence brightness of the chemosensory array in *P. aeruginosa* wild-type and several mutant strains.

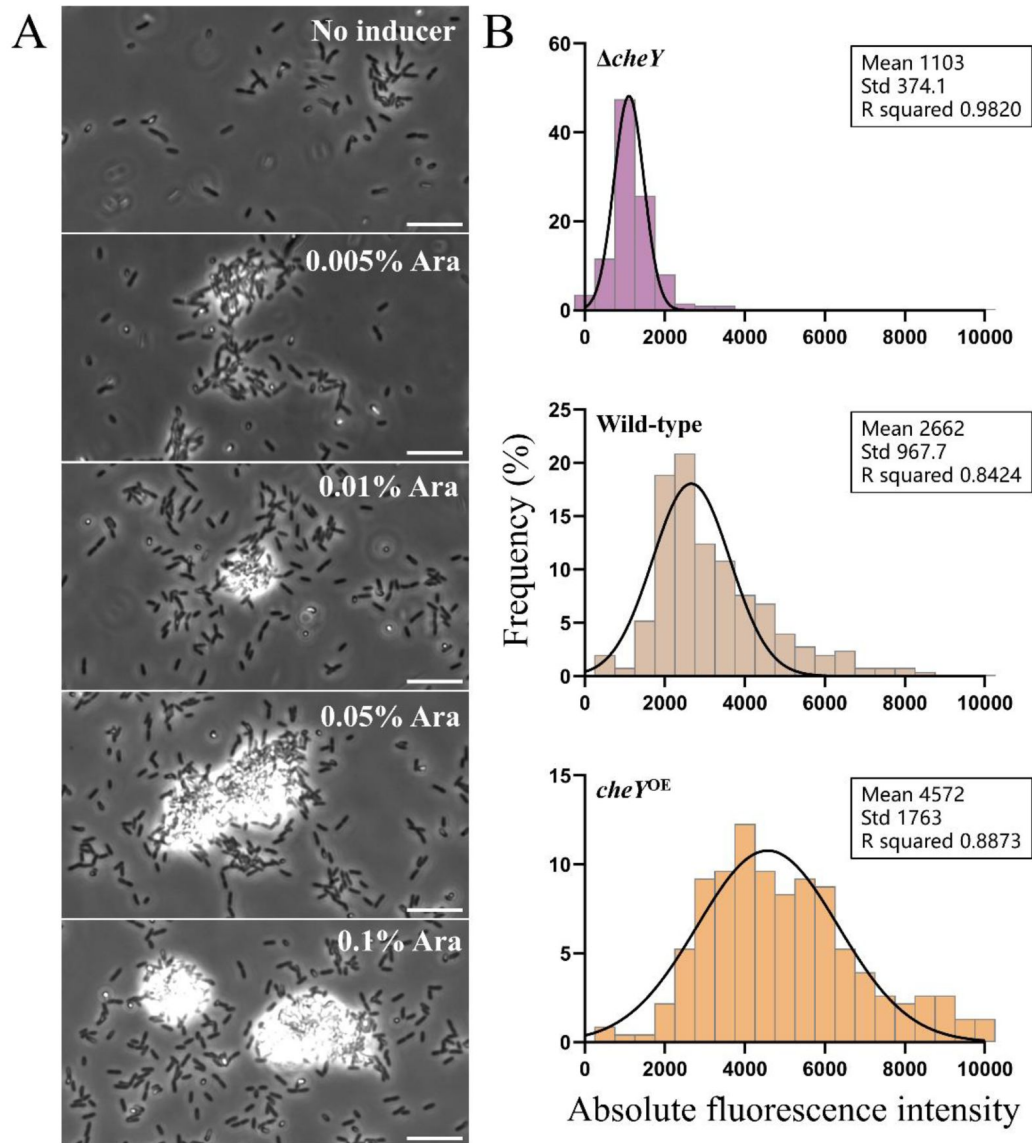


Fig. 4.

A. The evolution of cell aggregation as the intracellular CheY concentration increases by induction with higher concentrations of arabinose. The scale bar is 10 μ m. **B.** Quantitative characterization of intracellular c-di-GMP levels at different CheY concentrations. From top to bottom, they correspond to the $\Delta cheY$ strain (N=198), wildtype strain (N=249) and CheY overexpression strain (N=228), respectively.

Therefore, the colocalization of chemosensory array and flagellar motor facilitates the precise regulation of cell swimming direction within the low intracellular CheY-P concentration threshold, using locally distributed CheY-P molecules. This helps to avoid the occurrence of the aforementioned cross-pathway interference.

Summary and Discussion

P. aeruginosa harbors multiple signal transduction systems that regulate flagella-mediated swimming motility (Che pathway), pili-mediated interface twitching motility (Pil pathway), and biofilm formation (Wsp pathway)¹⁶. These systems allow it to thrive in various ecological niches within complex external environments. Here, employing a combination of chromosomal fluorescent protein fusion and flagellar labeling techniques in living cells, we directly observed that the chemosensory array of *P. aeruginosa* Che pathway and the flagellar motor share a high degree of consistency in their assembly sites. We found that the chemosensory array, comprising Che proteins, was consistently located at the cell pole where the flagellum was positioned. Based on these observations, we deduced the construction mode of the chemotactic network and flagellar motor during the entire cell growth cycle (Fig. 5A). As the cell body matures and elongates, a new flagellum grows at the opposing cell pole, accompanied by the assembly of a fresh chemosensory array. The cell then undergoes division from the middle, generating two daughter cells, each equipped with fully functional motility and chemotaxis capabilities. This ensures robust inheritance of these chemotaxis and motility-related macromolecular machines.

Based on our understanding of the regulation mechanism of flagellar position in *P. aeruginosa*, we constructed a $\Delta flhF$ mutant and found that the position distribution of its chemosensory array is actively regulated by FlhF-mediated molecular mechanisms, rather than being passively restricted by membrane curvature factors as seen in *E. coli*. A similar active regulation phenomenon was previously reported in *Vibrio cholerae*, with the corresponding molecular regulation mechanism detailed extensively²⁸. The flagella and chemosensory arrays of newborn *V. cholerae* cells are located at the old cell pole, with the polar anchor protein HubP distributed at both cell poles. As the cell grows, HubP recruits the ParA homologue ParC. Subsequently, ParC recruits ParP to assemble a new chemosensory array at the new cell pole. Upon cell division, HubP relocates to the middle of the cell, ensuring its robust presences at both poles of the daughter cells. Here, HubP recruits ParA homologues including FlhG, which influence flagellar motor assembly. However, previous experiments confirmed that the cell cycle-dependent polar localization of ParC and chemotactic proteins is independent of the flagellar position regulatory protein FlhF. This suggests that the localization of flagellar motors and chemosensory arrays in *V. cholerae* is controlled separately through different pathways. In contrast, flagellar motors were no longer robustly distributed at the cell pole after *flhF* was knocked out in *P. aeruginosa*, whereas the assembly site of chemosensory array and flagellar motor remained remarkably consistent. This suggests that FlhF regulates the distribution patterns of both and implies an interaction between them.

We further constructed $\DeltafliG$ and $\DeltafliF$ mutants, which disrupted the assembly of the flagellar motor. Surprisingly, we observed that the assembly of chemosensory array was impaired under these conditions, with a significant decrease in the proportion of individuals exhibiting obvious Che protein fluorescent bright spots. In contrast, the chemosensory array assembly in the stator mutant ($\Delta motA \Delta motCD$) was similar to that of the wild-type strain. This suggests that the integrity rather than functionality of the flagellar motor in *P. aeruginosa* is critical for the assembly of its chemosensory array, and its assembly efficiency is greatly reduced in the absence of flagellar motors. To eliminate the potential confounding effects of FlhF and flagellar motors, we constructed a $\Delta flhF \DeltafliF$ mutant strain. Under this condition, the proportion of individuals with obvious Che protein fluorescent bright spots dropped to ~50%, indicating that passive regulation based on membrane curvature may still exist in *P. aeruginosa*, albeit to a lesser degree.

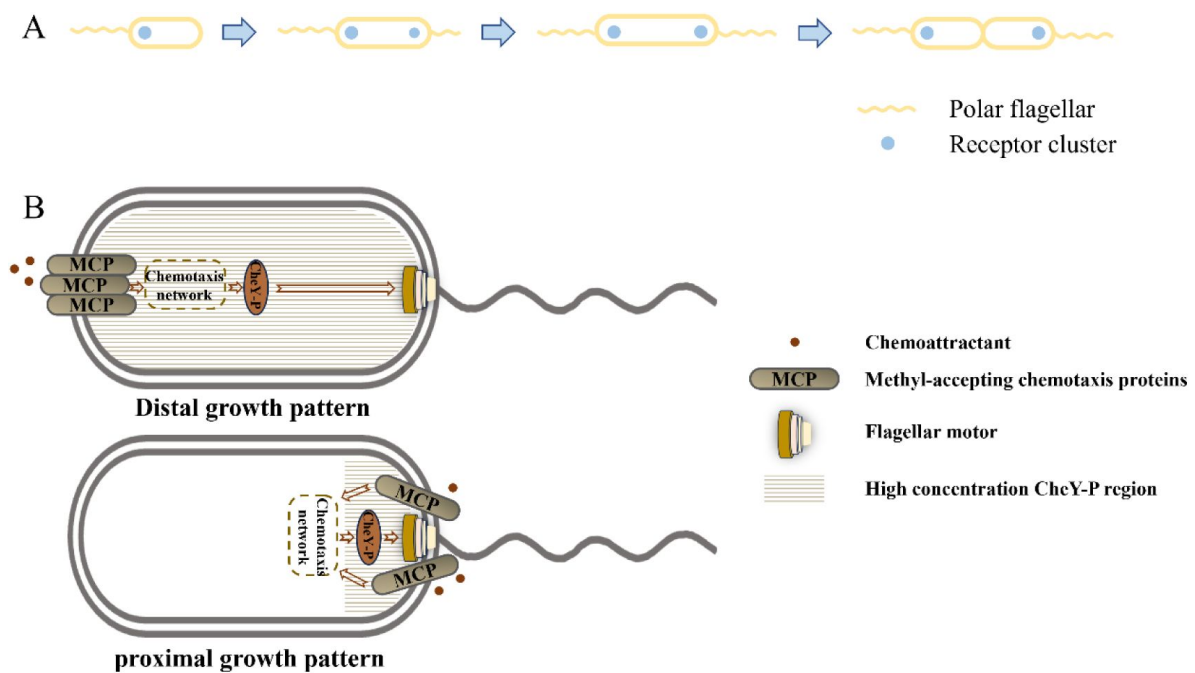


Fig. 5.

A. The construction mode of the chemotaxis network and flagellar motor throughout the complete cell growth cycle. **B.** The proximal growth mode of the *P. aeruginosa* flagellar motor and receptor clusters will effectively regulate the spatial range of CheY action, thus avoiding unintentional cross-pathway regulation.

Is the remarkable consistency in the spatial distribution of these two independent structural units an unintentional outcome or a hidden mystery? By introducing a c-di-GMP monitoring system at the single-cell level, we discovered that overexpressing CheY substantially increases the intracellular c-di-GMP concentration. Previous studies generally believed that CheY-P, as a chemotactic regulatory protein, only influences flagellar motor switching. Here, we found that CheY-P, as a signaling molecule, can accomplish threshold-limited cross-pathway regulation. At low CheY-P concentrations, it is conservatively involved in motor switching regulation, as confirmed by previous studies. At high CheY-P concentrations, cell motility is suppressed, and c-di-GMP is triggered to express, though its specific mechanism needs further exploration. The intracellular environment is highly crowded⁴³, and the proximal growth pattern shown in **Fig. 5B** has multiple physiological significances. On one hand, chemotaxis-related protein can precisely regulate cell motility with minimal synthesis costs. On the other hand, the chemotactic network and the flagellar motor are spatially integrated within the cell, ensuring their robust existence within the cell body and preventing mutual interference with collateral pathways.

Many bacteria possess multiple signal transduction pathways. The orderly operation of each pathway within the micrometer-scale space presents a fascinating scientific problem. Here, we utilized the well-known chemotaxis network as a starting point to shed light on this problem, providing inspiration for future in-depth understanding of related issues.

Materials and methods

Strains and cell culture

The strains and plasmids used in this study are listed in **Table 1**. The *Escherichia coli* TOP10 strain was used for standard genetic manipulations. A single-colony isolate was grown in 3 ml of LB broth (1% Bacto tryptone, 0.5% yeast extract and 1% NaCl) overnight to saturation on a rotary shaker (250 rpm) at 37°C. An aliquot was diluted 1:100 into 10 ml of LB broth and grew to exponential phase. Appropriate antibiotics were added if necessary to prevent plasmid loss: for *E. coli*, 15 µg/ml Gentamicin and 25 µg/ml Tetracycline; for *P. aeruginosa*, 30 µg/ml Gentamicin and 50 µg/ml Tetracycline. To induce protein expression, Isopropyl β-D-1-thiogalactopyranoside (IPTG, 1.0 mM) was added to strains with pME6032-derivative vectors, and arabinose (0.005%, 0.01%, 0.05%, 0.1%) was added to strains with pJN105-derivative vectors. 2 ml of cells were harvested by centrifugation at 2000×g for 2 min, washed twice in an equal volume of motility buffer (MB) [50 mM potassium phosphate, 15 µM EDTA, 0.15 M NaCl, 5 mM Mg²⁺ and 10 mM lactic acid (pH 7.0)], and resuspended in 5 ml MB for subsequent fluorescence observation.

Construction of in-frame deletion mutants

We used polymerase chain reaction (PCR) to generate ~1000 bp DNA fragments with upstream (Up) and downstream (Dn) sequences flanking the target gene (including *flhF*, *flhF*, *motA* and *flhG*) to be deleted. The Up and Dn DNA sequences were fused with the linearized pex18gm vector using Gibson assembly⁴⁵. The resulting vectors were transformed into *P. aeruginosa* by electroporation, and the desired knockout mutants were obtained by double selections on gentamicin plates (LB plates with 30 µg/ml gentamicin) and sucrose plates (NaCl-free LB plates with 15% sucrose) at 37°C⁴⁶.

Generation of chromosomal fusions of *yfp* to *P. aeruginosa* gene

To generate *eyfp* fusion at the N-terminus of *cheY*, we first used PCR to generate upstream and downstream DNA sequences flanking the *cheY* gene. A subsequent PCR reaction aimed to yield the *cheY-eyfp* fusion with a 3× glycine linker (GGCGGAGGA), with pVS88 serving as the source of the

Strain, Plasmid	Genotype, phenotype, and Description	Source
Strains		
<i>P. aeruginosa</i>		
PAO1	wild-type strain	Fan Jin Group
PAO1 <i>fliC</i> ^{T394C}	Replacement of chromosomal <i>fliC</i> in PAO1	Ref 31
PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i>	<i>yfp</i> fusions at the N-terminus of <i>cheY</i> in PAO1 <i>fliC</i> ^{T394C}	This work
PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i> Δ <i>flhF</i>	Nonpolar <i>flhF</i> deletion in PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i>	This work
PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i> Δ <i>fliF</i>	Nonpolar <i>fliF</i> deletion in PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i>	This work
PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i> Δ <i>fliG</i>	Nonpolar <i>fliG</i> deletion in PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i>	This work
PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i> Δ <i>flhF</i> Δ <i>fliF</i>	Nonpolar <i>fliF</i> deletion in PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i> Δ <i>flhF</i>	This work
PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i> Δ <i>motA</i> Δ <i>motCD</i>	Nonpolar <i>motAB</i> and <i>motCD</i> deletion in PAO1 <i>fliC</i> ^{T394C} <i>cheY-eyfp</i>	This work
<i>E. coli</i>		
Top10	F- <i>mcrA</i> Δ (<i>mrr-hsRMS-mcrBC</i>) Φ 80/ <i>lacZ</i> Δ <i>M15</i> Δ <i>lacX74</i> <i>recA1</i> <i>araD139</i> Δ (<i>araleu</i>) 7697 <i>galU</i> <i>galK</i> <i>rpsL</i> (<i>Nal</i> ^R) <i>endA1</i> <i>mupG</i>	Invitrogen
Plasmids		
pex18gm	oriT ⁺ <i>sacB</i> ⁺ ; gene replacement vector with MCS from pUC18; Gm ^r	Fan Jin Group
<i>flhF</i> -pex18gm	In-frame deletion of <i>flhF</i> cloned into pex18gm; Gm ^r	This work
<i>fliF</i> -pex18gm	In-frame deletion of <i>fliF</i> cloned into pex18gm; Gm ^r	This work
<i>fliG</i> -pex18gm	In-frame deletion of <i>fliG</i> cloned into pex18gm; Gm ^r	This work
<i>motA</i> -pex18gm	In-frame deletion of <i>motA</i> cloned into pex18gm; Gm ^r	This work
<i>motCD</i> -pex18gm	In-frame deletion of <i>motCD</i> cloned into pex18gm; Gm ^r	Ref 30
<i>cheY-eyfp</i> -pex18gm	<i>yfp</i> fusions <i>cheY</i> cloned into pex18gm; Gm ^r	This work
<i>cheY</i> -pJN105	<i>cheY</i> overexpression vector in pJN1005, <i>cheY</i> expression is controlled by P _{BAD} promoter; Gm ^r	This work
<i>cheY</i> -pME6032	<i>cheY</i> overexpression vector in pME6032, <i>cheY</i> expression is controlled by P _{lac} promoter; Tet ^r	This work
pCdrA- <i>gfp</i>	pUCP22-NotI based cyclic di-GMP level reporter, Gm ^r	Ref 42

Table 1.

Strains and plasmids used in this study.

enhanced yellow fluorescent (*eyfp*). The three DNA sequences, along with the linearized vector, were fused to *cheY-eyfp-pex18gm* using Gibson assembly. Finally, gene replacement was used to transfer the fusion constructs into the *P. aeruginosa* chromosome by homologous recombination.

Flagellar staining and fluorescence imaging

Flagellar filaments were labeled by following the protocol described previously⁴⁷. Cells (1 ml of exponential-phase culture) were harvested by centrifugation at 2000×*g* for 10 min and washed twice in 1 ml MB. The final pellet was adjusted to a volume of ~100 µl that concentrated the bacterial 10-fold. Alexa Fluor 568 maleimide (Invitrogen-Molecular Probes) was added to a final concentration of 20 µg/ml, and labeling was allowed to proceed for 30 min at room temperature with gyration at 80 rpm. Unused dye was then removed by washing the cells with MB three times, and the final pellet was resuspended in MB.

For fluorescence imaging, 50 µl of cells were added to the chamber (constructed with two double-sticky tapes, a glass slide, and a poly-L-Lysine coated coverslip), incubated for 3 min, and rinsed with 100 µl MB. The boundary of the chamber was then sealed with Apiezon vacuum grease. The chamber was placed on a Nikon Ti-E inverted fluorescence microscope with a 100× oil-immersion objective and a sCMOS camera (Primer95B, Photometrics). The flagellar filament and chemosensory array of cells were observed separately with corresponding filter-set, using a 200-ms exposure time. To quantify the fluorescence intensity of the receptor cluster, we took the intensity maximum point as the center and extract a 3×3 pixel matrix around it. The mean value of the elements in this matrix represents the final fluorescence intensity.

Western Blotting

The total protein of *P. aeruginosa* strain cells in each treatment group was extracted and its concentration was measured with Bicinchoninic acid (BCA Protein Quantification Kit, Yeasen Biotechnology Co., Ltd) when the cells grew to exponential phase with same reduced turbidity (OD₆₀₀ between 0.9 and 1.0). The exact amount of protein was subjected to SDS-PAGE electrophoresis, then transferred to a nitrocellulose membrane, which was blocked with 50 mg/mL skim milk in TBST buffer (20 mM Tris, 150 mM NaCl [pH7.4], 0.1% Tween 20) for 1 h at room temperature. Since the N-terminal sequence of EYFP is almost identical to that of GFP, a rabbit anti-GFP antibody (1:2000, Yeasen Biotechnology Co., Ltd) and a Mouse anti-β-actin antibody (1:2000, Invitrogen) were added and incubated at 4°C overnight. The nitrocellulose membrane was gently washed with TBST for 20 min, three times. Subsequently, HRP labeled goat anti-rabbit IgG (1:4000, Abcam) and rabbit anti-mouse IgG (1:4000, Abcam) were added and incubated at room temperature for 1 h. The nitrocellulose membrane was again gently washed with TBST for 20 min, three times. Then, the membrane was developed using the ECL chemiluminescence detection system (Beyotime P0018FS) and visualized with the ChemiDoc XRS+ system (Bio-Rad). Three independent experiments were conducted for reproducibility.

Monitoring c-di-GMP signal at the single-cell level

For measurements of c-di-GMP signaling, the validated reporter plasmid pCdrA-*gfp*, which produces green fluorescent protein (GFP) in response to an increase in c-di-GMP, was introduced into the experimental system. A 1000× diluted culture was inoculated into the apparatus described in the previous section and allowed to adhere to the surface for ~10 min, with the slight difference that the coverslips were not coated with poly-L-Lysine to avoid interference from background fluorescence. Bacteria were imaged using a Nikon Ti-E inverted microscope, which was equipped with an EMCCD camera (DU897, iXon3, Andor Technology). We used a 100× oil-immersion objective. Bacteria were illuminated with a 488-nm laser (Sapphire 488–200 mW, Coherent), using a standard GFP filter sets in the fluorescence imaging system with an exposure time of 500 ms. To

ensure the consistency of culture and observation conditions, the data of wild-type and *cheY*-pME6032 carried strains were collected at the same time. More than 200 independent individuals of each strain were used for data analysis.

Single-cell c-di-GMP concentration was characterized by calculating fluorescence intensity. The average fluorescence intensity of a single cell was calculated by dividing the total fluorescence intensity of a single cell by the cell volume. Bacteria were simplified to a hollow cylindrical trunk with hollow hemispherical caps at both ends. The cell volume can be calculated by taking the length of the short axis of the cell body as the diameter of the sphere and the length of the long axis as the sum of the height of the cylinder and the diameter of the sphere. The mean fluorescence intensity of the cell was obtained by subtracting the average background fluorescence intensity from the total fluorescence intensity of the cell, and then dividing by the cell volume.

Author Contributions

J.Y. and R.Z. designed the work; Z.W., M.T. and S.F. performed the measurements with help from M.C.; Z.W. and S.F. analyzed the data; Z.W., J.Y. and R.Z. wrote the paper. Z.W., M.T. and S.F. contributed equally to this work.

Conflicts of interest

The authors declare no competing interests.

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Editors

Reviewing Editor

Ariel Amir

Weizmann Institute of Science, Rehovot, Israel

Senior Editor

Dominique Soldati-Favre

University of Geneva, Geneva, Switzerland

Reviewer #1 (Public Review):

Summary:

The study by Wu et al presents interesting data on bacterial cell organization, a field that is progressing now, mainly due to the advances in microscopy. Based mainly on fluorescence microscopy images, the authors aim to demonstrate that the two structures that account for bacterial motility, the chemotaxis complex and the flagella, colocalize to the same pole in *Pseudomonas aeruginosa* cells and to expose the regulation underlying their spatial organization and functioning.

Strengths:

The subject is of importance.

Weaknesses:

The conclusions are too strong for the presented data. The lack of statistical analysis makes this paper incomplete. The novelty of the findings is not clear.

Major issues:

(1) The novelty is in question since in the Abstract the authors highlight their main finding, which is that both the chemotaxis complex and the flagella localize to the same pole, as surprising. However, in the Introduction they state that "pathway-related receptors that mediate chemotaxis, as well as the flagellum are localized at the same cell pole^{17,18}". I am not a *pseudomonas* researcher and from my short glance at these references, I could not tell whether they report colocalization of the two structures to the same pole. However, I trust the authors that they know the literature on the localization of the chemotaxis complex and flagella in their organism. See also major issue number 5 on the novelty regarding the involvement of c-di-GMP.

(2) Statistics for the microscopy images, on which most conclusions in this manuscript are based, are completely missing. Given that most micrographs present one or very few cells, together with the fact that almost all conclusions depend on whether certain macromolecules are at one or two poles and whether different complexes are in the same pole, proper statistics, based on hundreds of cells in several fields, are absolutely required. Without this information, the results are anecdotal and do not support the conclusions. Due to the importance of statistics for this manuscript, strict statistical tests should be used and reported. Moreover, representative large fields with many cells should be added as supportive information.

The problem is more pronounced when the authors make strong statements, as in lines 157-158: "The results revealed that the chemoreceptor arrays no longer grow robustly at the cell pole (Figure 2A)". Looking at the seven cells shown in Figure 2A, five of them show polar localization of the chemoreceptors. The question is then: what is the percentage of cells that

show precise polar, near-polar, or mid cell localization (the three patterns shown here) in the mutant and in the wild type? Since I know that these three patterns can also be observed in WT cells, what counts is the difference, and whether it is statistically significant.

Even for the graphs shown in Figures 3C and 3D, where the proportion of cells with obvious chemoreceptor arrays and absolute fluorescence brightness of the chemosensory array are shown, respectively, the questions that arise are: for how many individual cells these values hold and what is the significance of the difference between each two strains?

(3) The authors conclude that "Motor structural integrity is a prerequisite for chemoreceptor self-assembly" based on the reduction in cells with chemoreceptor clusters in mutants deleted for flagellar genes, despite the proper polar localization of the chemotaxis protein CheY. They show that the level of CheY in the WT and the mutant strains is similar, based on Western blot, which in my opinion is over-exposed. "To ascertain whether it is motor integrity rather than functionality that influences the efficiency of chemosensory array assembly", they constructed a mutant deleted for the flagella stator and found that the motor is stalled while CheY behaves like in WT cells. The authors further "quantified the proportion of cells with receptor clusters and the absolute fluorescence intensity of individual clusters (Figures 3C-D)". While Figure 3DC suggests that, indeed, the flagella mutants show fewer cells with a chemotaxis complex, Figure 3D suggests that the differences in fluorescence intensity are not statistically significant.

Since it is obvious that the regulation of both structures' production and localization is codependent, I think that it takes more than a Western blot to make such a decision.

(4) I wonder why the authors chose to label CheY, which is the only component of the chemotaxis complex that shuttles back and forth to the base of the flagella. In any case, I think that they should strengthen their results by repeating some key experiments with labeled CheW or CheA.

(5) The last section of the results is very problematic, regarding the rationale, the conclusions, and the novelty. As far as the rationale is concerned, I do not understand why the authors assume that "a spatial separation between the chemoreceptors and flagellar motors should not significantly impact the temporal comparison in bacterial chemotaxis". Is there any proof for that? More surprising for me was to read that "The signal transduction pathways in *E. coli* are relatively simple, and the chemotaxis response regulator CheY-P affects only the regulation of motor switching". There are degrees of complexity among signal transduction pathways in *E. coli*, but the chemotaxis seems to be ranked at the top. CheY is part of the adaptation. Perfect adaptation, as many other issues related to the chemotaxis pathway, which include the wide dynamic range, the robustness, the sensitivity, and the signal amplification (gain), are still largely unexplained. Hence, such assumptions are not justified.

More perplexing is the novelty of the authors' documentation of the effect of the chemotaxis proteins on the c-di-GMP level. In 2013, Kulasekara et al. published a paper in eLife entitled "c-di-GMP heterogeneity is generated by the chemotaxis machinery to regulate flagellar motility". In the same year, Kulasekara published a paper entitled "Insight into a Mechanism Generating Cyclic di-GMP Heterogeneity in *Pseudomonas aeruginosa*". The authors did not cite these works and I wonder why.

(6) Throughout the manuscript, the authors refer to foci of fluorescent CheY as "chemoreceptor arrays". If anything, these foci signify the chemotaxis complex, not the membrane-traversing chemoreceptors.

Conclusions:

The manuscript addresses an interesting subject and contains interesting, but incomplete, data.

Reviewer #2 (Public Review):**Summary:**

Here, the authors studied the molecular mechanisms by which the chemoreceptor cluster and flagella motor of *Pseudomonas aeruginosa* (PA) are spatially organized in the cell. They argue that FlhF is involved in localizing the receptors-motor to the cell pole, and even without FlhF, the two are colocalized. FlhF is known to cause the motor to localize to the pole in a different bacterial species, *Vibrio cholera*, but it is not involved in receptor localization in that bacterium. Finally, the authors argue that the functional reason for this colocalization is to insulate chemotactic signaling from other signaling pathways, such as cyclic-di-GMP signaling.

Strengths:

The experiments and data look to be high-quality.

Weaknesses:

However, the interpretations and conclusions drawn from the experimental observations are not fully justified in my opinion.

I see two main issues with the evidence provided for the authors' claims.

(1) Assumptions about receptor localization:

The authors rely on YFP-tagged CheY to identify the location of the receptor cluster, but CheY is a diffusible cytoplasmic protein. In *E. coli*, CheY has been shown to localize at the receptor cluster, but the evidence for this in PA is less strong. The authors refer to a paper by Guvener et al 2006, which showed that CheY localizes to a cell pole, and CheA (a receptor cluster protein) also localizes to a pole, but my understanding is that colocalization of CheY and CheA was not shown. My concern is that CheY could instead localize to the motor in PA, say by binding FlhM. This "null model" would explain the authors' observations, without colocalization of the receptors and motor.

Verifying that CheY and CheA are colocalized in PA would be a very helpful experiment to address this weakness.

(2) Argument for the functional importance of receptor-motor colocalization at the pole:

The authors argue that colocalization of the receptors and motors at the pole is important because it could keep phosphorylated CheY, CheY-p, restricted to a small region of the cell, preventing crosstalk with other signaling pathways. Their evidence for this is that overexpressing CheY leads to higher intracellular c-di-GMP levels and cell aggregation.

Say that the receptors and motors are colocalized at the pole. In *E. coli*, CheY-p rapidly diffuses through the cell. What would prevent this from occurring in PA, even with colocalization?

Elevating CheY concentration may increase the concentration of CheY-p in the cell, but might also stress the cells in other unexpected ways. It is not so clear from this experiment that elevated CheY-p throughout the cell is the reason that they aggregate, or that this outcome is avoided by colocalizing the receptors and motor at the same pole.

If localization of the receptor array and motor at one pole were important for keeping CheY-p levels low at the opposite pole, then we should expect cells in which the receptors and motor are not at the pole to have higher CheY-p at the opposite pole. According to the authors' argument, it seems like this should cause elevated cdG levels and aggregation in the delta flhF mutants with wild-type levels of CheY. But it does not look like this happened.

Instead of varying CheY expression, the authors could test their hypothesis that receptor-motor colocalization at the pole is important for preventing crosstalk by measuring cdG levels in the flhF mutant, in which the motor (and maybe the receptor cluster) are no longer localized in the cell pole.

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Reviewer #3 (Public Review):

Summary:

The authors investigated the assembly and polar localization of the chemosensory cluster in *P. aeruginosa*. They discovered that a certain protein (FlhF) is required for the polar localization of the chemosensory cluster while a fully-assembled motor is necessary for the assembly of the cluster. They found that flagella and chemosensory clusters always co-localize in the cell; either at the cell pole in wild-type cells or randomly-located in the cell in FlhF mutant cells. They hypothesize that this co-localization is required to keep the level of another protein (CheY-P), which controls motor switching, at low levels as the presence of high levels of this protein (if the flagella and chemosensory clusters were not co-localized) is associated with high-levels of c-di-GMP and cell aggregations.

Strengths:

The manuscript is clearly written and straightforward. The authors applied multiple techniques to study the bacterial motility system including fluorescence light microscopy and gene editing. In general, the work enhances our understanding of the subtlety of interaction between the chemosensory cluster and the flagellar motor to regulate cell motility.

Weaknesses:

The major weakness in this paper is that the authors never discussed how the flagellar gene expression is controlled in *P. aeruginosa*. For example, in *E. coli* there is a transcriptional hierarchy for the flagellar genes (early, middle, and late genes, see Chilcott and Hughes, 2000). Similarly, *Campylobacter* and *Helicobacter* have a different regulatory cascade for their flagellar genes (See Lertsethtakarn, Ottemann, and Hendrixson, 2011). How does the expression of flagellar genes in *P. aeruginosa* compare to other species? How many classes are there for these genes? Is there a hierarchy in their expression and how does this affect the results of the FliF and FliG mutants? In other words, if FliF and FliG are in class I (as in *E. coli*) then their absence might affect the expression of other later flagellar genes in subsequent classes (i.e., chemosensory genes). Also, in both FliF and FliG mutants no assembly intermediates of the flagellar motor are present in the cell as FliG is required for the assembly of FliF (see Hiroyuki Terashima et al. 2020, Kaplan et al. 2019, Kaplan et al. 2022). It could be argued that when the motor is not assembled then this will affect the expression of the other genes (e.g., those of the chemosensory cluster) which might play a role in the decreased level of chemosensory clusters the authors find in these mutants.

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